25900013 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBIF100 - BIOLOGICAL DATA AND MANAGEMENT

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Describe two advantages of using a Database Management System (DBMS). [U] / [CO1]
2. Articulate two features of OMIM. [A] / [CO4]
3. Demonstrate nature of biological data. [U] / [CO2]
4. Discuss classification of biological database. [U] / [CO2]
5. Prepare a list of Literature databases. [A] / [CO4]
6. Discover the role of DDBJ in sequence storage. [A] / [CO4]
7. Report an interesting feature of Prosite. [A] / [CO4]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

8. Compare relational databases and NoSQL databases. [U] / [CO1]

9. Compare and contrast physical and logical database schemas. [U] / [CO1]

10. Describe the historical development of biological databases. [U] / [CO2]

11. Discuss how data collected and stored in primary databases [U] / [CO3]

12. Illustrate a SWISS-PROT entry format. [A] / [CO4]

13. Examine the features of a genome database entry. [A] / [CO4]

[5x4 = 20]

Part C

Long essay Questions

Answer any **2** questions

Each question carries **10** marks

14. Relate the importance of data consistency, isolation, and durability in a database transaction. Illustrate with examples. [U] / [CO1]

15. Discuss how data trends have evolved over the years and their impact on decision-making in various industries. [U] / [CO1]

16. Discuss the significance of NCBI in bioinformatics and its contributions to biological research. [U] / [CO2]

17. Articulate the importance of compound databases. [A] / [CO4]

[10x2 = 20]